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Studierichting: Informatica
Oefeningenreeks 6E nr. 1.6

$T_4\left(f,\frac{\pi}{2}\right)(x)\text{voor}f(x)=\sin x$

Eerste 4 afgeleiden:

$f(x)=\sin x$

$f'(x)=\cos x$

$f''(x)=-\sin x$

$f'''(x)=-\cos x$

$f^{iv}(x)=\sin x$

$$T_4\left(f,\frac{\pi}{2}\right)(x)=\sin\frac{\pi}{2}+\cos\frac{\pi}{2}\left(x-\frac{\pi}{2}\right)-\frac{\sin\frac{\pi}{2}}{2!}\left(x-\frac{\pi}{2}\right)^2-\frac{\cos\frac{\pi}{2}}{3!}\left(x-\frac{\pi}{2}\right)^3+\frac{\sin\frac{\pi}{2}}{4!}\left(x-\frac{\pi}{2}\right)^4$$

$$=1+0-\frac{1}{2}\left(x-\frac{\pi}{2}\right)^2-0+\frac{1}{24}\left(x-\frac{\pi}{2}\right)^4$$

$$=1-\frac{1}{2}\left(x^2-\pi x+\frac{\pi^2}{4}\right)+\frac{1}{24}\left(x^2-\pi x+\frac{\pi^2}{4}\right)^2$$

$$=1-\frac{1}{2}\left(x^2-\pi x+\frac{\pi^2}{4}\right)+\frac{1}{24}\left(x^4-2\pi x^3+\frac{6\pi^2x^2}{4}-\frac{2\pi^3x}{4}+\frac{\pi^4}{16}\right)$$

$$=1-\frac{x^2}{2}+\frac{\pi x}{2}-\frac{\pi^2}{8}+\frac{x^4}{24}-\frac{\pi x^3}{12}+\frac{\pi^2x^2}{16}-\frac{\pi^3x}{48}+\frac{\pi^4}{384}$$

$$=1-\frac{\pi^2}{8}+\frac{\pi^4}{384}+\left(\frac{\pi}{2}-\frac{\pi^3}{48}\right)x+\left(\frac{\pi^2}{16}-\frac{1}{2}\right)x^2-\frac{\pi x^3}{12}+\frac{x^4}{24}$$

References